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The Mechanics of the Industrial Revolution

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1 Overview

- **Question.** Why did Industrial Revolution manufacturing (especially cotton textiles and metals) emerge where it did *within* Britain?
- **Core hypothesis.** Cross-county variation in *mechanical and toolmaking skills* (a very specific “artisan human capital” stock) is a key prerequisite for adopting and operating Industrial Revolution technologies.
- **Complementary forces.** Low pre-boom agricultural wages (proxying for low agricultural productivity) and market access matter as well; coal proximity matters more for metals than for cotton textiles.
- **Empirical design.** County-level (England and Wales) cross-section regressions linking pre-Industrial characteristics (circa 1700–1790) to later industrial employment (1831, 1851).
- **Methodological emphasis.** Spatial correlation is treated as an *omitted-variable / confounding* problem, not just a standard-error problem. The main workhorse is a semiparametric spatial regression that flexibly controls for geography.

2 Conceptual mechanism (Fig. 1)

The paper’s narrative mechanism is:

1. **Market integration** lowers the relative price of agricultural goods.
2. Counties with **poor soil / low agricultural productivity** (empirically proxied by *low agricultural wages*) lose comparative advantage in agriculture and shift toward manufacturing.
3. Only some manufacturing activities generate or rely on the **mechanical skills** needed for the Industrial Revolution (machinery, repairs, precision tools, etc.). Those counties are better positioned to adopt new technologies like mechanized spinning and weaving.

A compact way to organize the empirics is:

$$\text{Industrialization}_i \approx F\left(\underbrace{\text{MechanicalSkill}_i}_{\text{capability}}, \underbrace{\text{Wage}_i}_{\text{ag. productivity}}, \underbrace{\text{MarketPotential}_i}_{\text{demand/access}}, \underbrace{\text{CoalDistance}_i}_{\text{energy input}}, \underbrace{\text{WaterPower}_i}_{\text{early energy}}\right), \quad (1)$$

with comparative statics:

$$\frac{\partial \text{Industrialization}_i}{\partial \text{MechanicalSkill}_i} > 0, \quad \frac{\partial \text{Industrialization}_i}{\partial \text{Wage}_i} < 0, \quad \frac{\partial \text{Industrialization}_i}{\partial \text{MarketPotential}_i} > 0, \quad \frac{\partial \text{Metals}_i}{\partial \text{CoalDistance}_i} < 0.$$

Figure 1 interpretation

Takeaway: This is a *path dependence* story. The spatial distribution of pre-industrial manufacturing (and the skills it creates) shapes where later mechanized technologies can profitably locate.

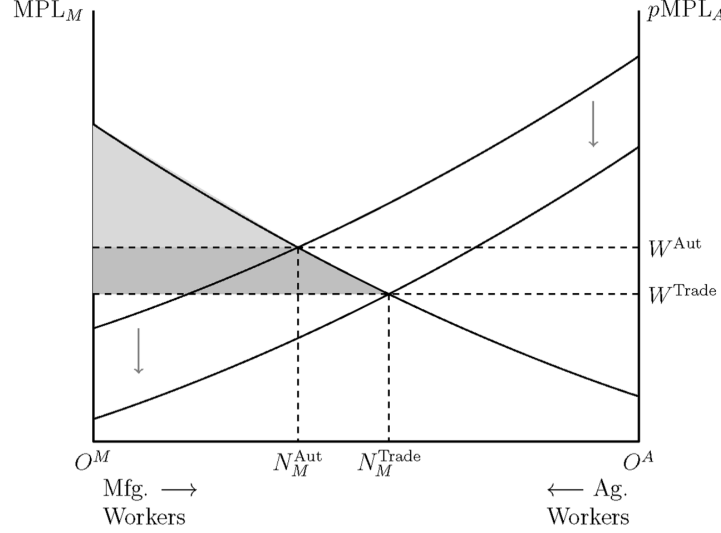


FIG. 1.—A fall in the relative price of agricultural goods caused by market integration leads areas with poor soil to specialize in various manufacturing activities. Some of these activities have the potential to generate new skills of the sort that underlay British industrialization.

3 Data, key variables, and construction

Unit of observation is an English/Welsh county ($N \approx 41$). The paper carefully uses *predetermined* covariates (mostly pre-1790) to explain later outcomes (1831/1851).

3.1 Outcomes

- **Textiles 1831:** share of men employed in textiles (county of residence).
- **Textiles 1851:** share of men (age 20–29) employed in textiles (county of residence).
- **Metallurgy 1851:** share of men employed in metallurgy (and subcategories).
- **Traditional industries 1851:** shares in food, garments, shoes, wood.

3.2 Core predictors

- **Mechanical skill (“Skills 1790s”).** From the 1851 census: for each county i ,

$$\text{Skill}_i \equiv \frac{\#\{\text{mechanics \& toolmakers aged 60+ in 1851, born in } i\}}{\#\{\text{all men aged 60+ in 1851, born in } i\}}. \quad (2)$$

Interpretation: men aged 60+ in 1851 were born around the 1790s, so their occupational structure proxies for the local skill supply around the onset of mechanization.

- **Agricultural wage 1760s.** A proxy for agricultural productivity/soil quality; used in logs.

- **Market potential 1750.** A gravity-style measure:

$$MP_i \equiv \sum_j \frac{\text{Income}_j}{d_{ij}^2}, \quad \text{Income}_j \equiv (\text{Wage}_{j, 1760s}) \times (\text{Population}_{j, 1750}), \quad (3)$$

where d_{ij} is distance between county centroids.

- **Coal distance.** $\text{CoalDist}_i = \min_{j \in \mathcal{C}} d_{ij}$, where $\mathcal{C}$ are counties with coalfields, with coal counties assigned a positive baseline distance (not zero), then logged in regressions.
- **Water power potential.** A geographic proxy constructed from drainage area and slope; in practice used in logs.

3.3 Maps and reduced-form visual evidence (Figs. 2–4)

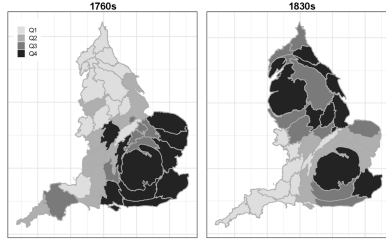


FIG. 2.—Cartograms of England in the 1760s and 1830s. The area of each county is scaled in proportion to its aggregate labor income (wage times population), and shaded according to its wage rate.

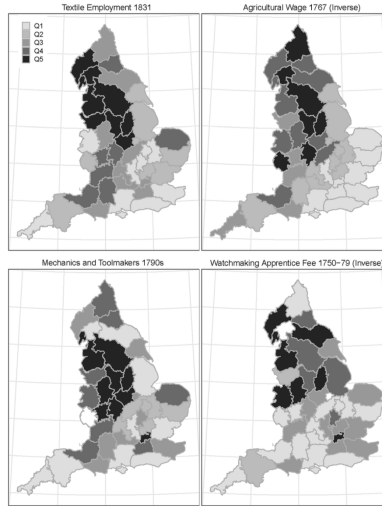


FIG. 3.—County levels, shaded by quintile, of textile employment in 1831, the inverse wage of agricultural laborers, the share of mechanics and toolmakers in the labor force in the 1790s, and the inverse cost of becoming a watchmaking apprentice. The paper predicts that successful textile areas in the nineteenth century will have had low wages and high levels of mechanical skill in the eighteenth century. The cost of becoming a watchmaking apprentice is used as an instrument to deal with potential issues of endogeneity and measurement error in the mechanics and toolmakers variable.

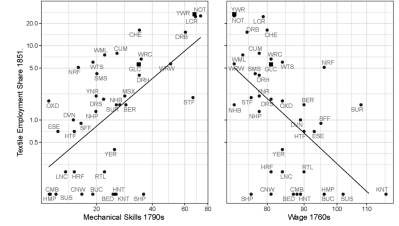


FIG. 4.—Supply of mechanical skill in the 1790s and agricultural wages in the 1760s versus the percentage of males employed in textiles in 1851; logarithmic axes. Note how the decline of Gloucestershire (GLC; the dominant woolen textile center in the mid-eighteenth century) and the rise of West Yorkshire (YWR) can be predicted by their respective supplies of mechanical skills in the 1790s. Full county names corresponding to these abbreviations can be found in the replication files.

4 Econometric models (the “models” in this paper)

4.1 (M1) Semiparametric spatial regression (main workhorse)

The core specification is a county-level cross-sectional regression with a flexible geographic control:

$$y_i = \alpha + X_i' \beta + f(\text{lon}_i, \text{lat}_i) + \varepsilon_i, \quad (4)$$

where:

- y_i is an outcome such as the textile employment share (1831 or 1851) or metallurgy share (1851),

- X_i includes log skills, log wages, log market potential, and optionally log distance to coal and log water power,
- $f(\cdot)$ is a smooth function (thin-plate spline) capturing broad spatial trends and unobserved geography.

A standard way to represent estimation is penalized least squares:

$$(\hat{\alpha}, \hat{\beta}, \hat{f}) = \arg \min_{\alpha, \beta, f} \sum_{i=1}^N \left(y_i - \alpha - X_i' \beta - f(\text{lon}_i, \text{lat}_i) \right)^2 + \lambda \mathcal{J}(f), \quad (5)$$

where $\mathcal{J}(f)$ penalizes the “wiggleness” of f (for thin-plate splines, a functional of second derivatives) and λ is chosen by a data-driven criterion (e.g., generalized cross-validation / AIC as in `mgcv`).

Interpretation of β . With outcomes in levels (employment shares) and covariates in logs, β_k is a *semi-elasticity*: a 1% increase in $X_{k,i}$ changes y_i by $\beta_k/100$ units of the dependent-variable scale (often interpretable as percentage points).

4.2 (M2) Skill supply regression

To study where mechanical skill comes from, the paper estimates variants of:

$$\text{SkillShare}_i = \alpha + \delta_1 \log(\text{ApprenticeFee}_i) + \delta_2 Z_i + f(\text{lon}_i, \text{lat}_i) + u_i, \quad (6)$$

where Z_i includes early population density, land tax intensity, distance to coal, and similar background covariates.

4.3 (M3) IV / 2SLS to address endogeneity or measurement error in skills

Mechanical skills may be endogenous and/or measured with error, so the paper uses watchmaking apprenticeship fees as an instrument.

First stage (conceptually):

$$\log(\text{Skill}_i) = \pi_0 + \pi_1 \log(\text{ApprenticeFee}_i) + \pi_2 W_i + f(\text{lon}_i, \text{lat}_i) + v_i, \quad \pi_1 < 0. \quad (7)$$

Second stage:

$$y_i = \alpha + \beta_1 \widehat{\log(\text{Skill}_i)} + \beta_2 W_i + f(\text{lon}_i, \text{lat}_i) + \varepsilon_i. \quad (8)$$

4.4 (M4) Spatial-confounding framework (Appendix A)

Appendix A formalizes the spatial issue as:

$$y = X\beta + \eta, \quad (9)$$

with spatial structure in η :

$$\eta = \tau(\text{lon}, \text{lat}) + u, \quad (10)$$

and spatial structure in regressors:

$$X_k = g_k(\text{lon}, \text{lat}) + Z_k. \quad (11)$$

If $g_k(\cdot)$ and $\tau(\cdot)$ are correlated, then (without controls) X_k is correlated with the regression error. The semiparametric approach (M1) is designed to absorb $\tau(\cdot)$ flexibly.

5 Main empirical results: textiles

5.1 Table 1: baseline determinants of textile employment (1831 and 1851)

Model. Equation (4) with y_i = textile employment share; X_i includes log skills (1790s), log wage (1760s), log market potential (1750), plus coal distance and water power in some columns.

Key patterns (Table 1).

- **Mechanical skill is strongly positive.** Higher mechanical skill predicts substantially more textile employment.
- **Wages are negative.** Lower agricultural wages (worse agricultural productivity) predict more later textiles.
- **Market potential is positive.** Better market access predicts more textiles.
- **Coal distance is not central for early textiles.** (In contrast, coal is more relevant for metals.)
- **Geographic confounding is addressed.** The flexible $f(\text{lon}, \text{lat})$ absorbs broad spatial trends; results are not just “north vs south.”

Reading: Skills look like a genuine capability constraint: even among low-wage counties, the ones with more mechanical skill endowment industrialize more in textiles.

5.2 Table 2: alternative channels (literacy, banks, legal services, information)

Model. Same as Table 1 but adds one candidate “institution/human capital” proxy at a time: literacy (around 1800), county banks (1796), lawyers (1730), booksellers (1761).

Reading.

- These variables do *not* robustly predict textiles once the main triad (skills, wages, market potential) and geography are controlled for.
- This pushes against a simple story where broad literacy or finance alone explains the county pattern of mechanized textiles.

TABLE 1
TEXTILE EMPLOYMENT IN 1831 AND 1851: SEMIPARAMETRIC SPATIAL REGRESSIONS
(41 Observations)

	DEPENDENT VARIABLE: SHARE OF MEN EMPLOYED IN TEXTILES					
	1831			1851		
	(1)	(2)	(3)	(4)	(5)	(6)
Skills, 1790s	2.022 (.503)	2.466 (.668)	2.020 (.511)	1.151 (.398)	1.079 (.532)	1.007 (.377)
Wage, 1760s	-4.776 (2.742)	-5.402 (2.817)	-4.783 (2.937)	-3.035 (2.109)	-3.004 (2.138)	-1.623 (2.056)
Market potential, 1750	.712 (.301)	.592 (.320)	.712 (.306)	.503 (.217)	.514 (.227)	.403 (.209)
Distance to coal		.720 (.685)			-.127 (.650)	
Water power			.001 (.425)			-.664 (.348)
Adjusted R^2	.684	.695	.676	.813	.810	.865
AIC	131.1	130.6	132.9	88.7	89.5	74.3
$S(\text{lon}, \text{lat})$	.641	.534	.768	.070	.076	.013

NOTE.—Dependent variable is the employment share of textiles among all males over 20 in 1831 and among males aged 20–29 in 1851. Semiparametric regressions with a thin-plate spline in longitude and latitude. $S(\text{lon}, \text{lat})$ gives the approximate significance. Standard errors are in parentheses. Regressions include a dummy in 1831 for Shropshire and a dummy in 1851 for Staffordshire. All other variables are in logs. AIC = Akaike information criterion.

TABLE 2
OTHER EXPLANATORY VARIABLES FOR TEXTILE EMPLOYMENT:
SEMIPARAMETRIC SPATIAL REGRESSIONS

	DEPENDENT VARIABLE: SHARE OF MEN EMPLOYED IN TEXTILES	
	1831	1851
Literacy, c. 1800	-1.671 (1.937)	-.140 (1.580)
County banks, 1796	-.064 (.431)	-.342 (.368)
Lawyers, 1730	.016 (.444)	-.378 (.386)
Booksellers, 1761	.023 (.358)	.155 (.294)
Adjusted R^2	.649	.652
AIC	137.6	114.1
$S(\text{lon}, \text{lat})$	.682	.709

NOTE.—Semiparametric spatial regressions. Dependent variable is the employment share of textiles among all males over 20 in 1831 and among males aged 20–29 in 1851. Standard errors are in parentheses. The approximate significance of the thin-plate spline is given by $S(\text{lon}, \text{lat})$. Regressions include skill, wages, market potential, and county dummies from first columns of table 1. “Literacy” is the percentage of convicts who were literate. “Density” is the density of 1700 population relative to county farmland. “County banks” is the number of county banks, “Booksellers” the number of booksellers, and “Lawyers” the number of attorneys, all per capita. AIC = Akaike information criterion.

6 Mechanism and endogeneity checks

6.1 Fig. 5 and Table 3: where do mechanical skills come from?

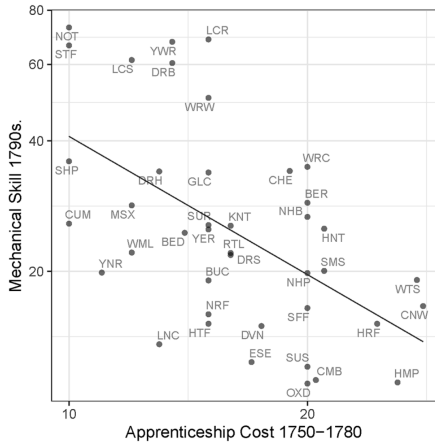


FIG. 5.—Apprenticeship fees for watchmaking 1750–80, and the supply of mechanical skill in the 1790s; logarithmic axes. Full county names corresponding to these abbreviations can be found in the replication files.

TABLE 3
FACTORS AFFECTING THE SUPPLY OF MECHANICAL SKILL:
SEMIPARAMETRIC SPATIAL REGRESSIONS

	DEPENDENT VARIABLE: SHARE OF MECHANICS IN 1851			
Apprentice fee	-.570 (.239)	-.491 (.164)	-.531 (.133)	
Apprentice fee/wage				-.475 (.138)
Land tax, 1707		-.085 (.141)	-.080 (.115)	-.083 (.120)
Population density, 1700		.818 (.144)	.552 (.133)	.533 (.139)
Distance to coal			-.596 (.140)	-.603 (.147)
Adjusted R^2	.533	.841	.898	.888
AIC	30.0	-13.0	-31.8	-27.7
$S(\text{lon}, \text{lat})$	.105	.004	.003	.005

NOTE.—Dependent variable is the share of males aged 60–69 listed as mechanics in 1851, by county of birth. Semiparametric spatial regressions: $S(\text{lon}, \text{lat})$ gives approximate significance level of a thin-plate spline in longitude and latitude. Standard errors are in parentheses. Regressions including population density include a dummy for London. All other variables are in logs. AIC = Akaike information criterion.

Interpretation of Fig. 5:

- A steep negative relationship between apprenticeship fees and later mechanical skill suggests a *supply-side* skill-formation channel: lower barriers/costs to training $\Rightarrow$ more skilled artisans.
- This also motivates the IV strategy: fees are a plausibly predetermined shifter of skill supply.

6.2 Tables 3–4: skill formation and SP/OLS/IV comparisons

Table 3 (skill formation).

- Apprentice fees (or fees relative to wages) negatively predict the mechanical-skill measure.
- This is consistent with a Becker-style human capital supply channel: higher training costs reduce skill acquisition.

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AIC	30.0	-13.0	-31.8	-27.7
S(lon, lat)	.105	.004	.003	.005

NOTE.—Dependent variable is the share of males aged 60–69 listed as mechanics in 1851, by county of birth. Semiparametric spatial regressions: S(lon, lat) gives approximate significance level of a thin-plate spline in longitude and latitude. Standard errors are in parentheses. Regressions including population density include a dummy for London. All other variables are in logs. AIC = Akaike information criterion.

TABLE 4
TEXTILE EMPLOYMENT IN 1831 AND 1851: SEMIPARAMETRIC, OLS, AND IV

	SHARE OF MEN EMPLOYED IN TEXTILES					
	1831			1851		
	SP	OLS	IV	SP	OLS	IV
Skills, 1790s	2.022 (.503)	2.125 (.459)	3.178 (1.422)	1.151 (.398)	1.703 (.332)	2.466 (1.119)
Wage, 1760s	-4.776 (2.742)	-6.960 (1.599)	-5.169 (2.634)	-3.035 (2.109)	-4.908 (1.136)	-3.686 (2.026)
Market potential, 1750	.712 (.301)	.713 (.228)	.484 (.481)	.503 (.217)	.499 (.188)	.340 (.392)
Adjusted R^2	.684	.655	.606	.813	.613	.567
Wu-Hausman	...	...	.286	...	...	.388
Weak instruments	...	...	.007	...	...	.015
S(lon, lat)	.641	...	...	.070	...	...

NOTE.—Dependent variable is the employment share of textiles among all males over 20 in 1831 and among males aged 20–29 in 1851. Apprentice fees are used as the instrument for mechanical skills. “SP” denotes spatial semiparametric regressions. Heteroskedasticity-robust standard errors are in parentheses. Regressions include longitude and latitude and a dummy in 1831 for Shropshire and in 1851 for Staffordshire. All other variables are in logs. “Wu-Hausman” and “Weak instruments” report the significance levels of tests for differences in OLS and IV coefficients and for weak instruments, respectively.

Table 4 (SP vs OLS vs IV).

- Skill coefficient is positive across SP, OLS, and IV.
- IV skill coefficients are larger, consistent with attenuation from measurement error and/or downward bias in OLS/SP.
- Diagnostics in this small sample suggest the instrument is not obviously weak, and Wu-Hausman does not strongly reject OLS (but power is limited).

7 Beyond textiles: metals vs traditional industries

7.1 Table 5 and Fig. 6: metallurgy (coal and persistence)

Model. Equation (4) with y_i = metallurgy employment share in 1851.

Interpretation.

- The “traditional metal area” indicator is strongly positive $\Rightarrow$ industrial clustering/persistence matters a lot for metals.
- Distance to coal enters with the expected *negative sign* in most specifications; Figure 6 shows the strong unconditional coal gradient.
- Mechanical skills are positively associated with metal employment (especially metal products), consistent with skill complementarity in metalworking.

7.2 Table 6: “placebo” traditional industries

Purpose. Check whether mechanical skills simply proxy for generic industrialization or county size.

Interpretation.

- In food, garments, shoes, and wood, mechanical skills are not strong predictors.

TABLE 5
DETERMINANTS OF EMPLOYMENT IN METALLURGY IN 1851

	DEPENDENT VARIABLE: EMPLOYMENT IN METALLURGY, 1851			
	Total	Metal Manufacturing	Metal Products	Sheet Metal
Traditional metal area	1.197 (.159)	1.966 (.411)	1.114 (.145)	.864 (.283)
Distance to coal	-.145 (.166)	-.185 (.293)	.039 (.121)	-.244 (.299)
Market potential, 1750	.139 (.065)	.558 (.188)	.097 (.059)	.383 (.114)
Skills, 1790s	.314 (.144)	.444 (.359)	.323 (.128)	.138 (.257)
Adjusted R^2	.874	.756	.819	.670
AIC	-4.8	78.5	-6.4	40.8
$S(\text{lon, lat})$	.071	.056	.077	.218

NOTE.—Dependent variable is the share of males aged 20–29 employed in metallurgy. Standard errors are in parentheses. Semiparametric regressions: $S(\text{lon, lat})$ is the approximate significance of a thin-plate spline term in longitude and latitude. “Traditional metal area” is a dummy for three West Midland counties: Staffordshire, Warwickshire, and Worcestershire. “Metal manufacturing” contains a dummy for Rutland. All other variables are in logs. AIC = Akaike information criterion.

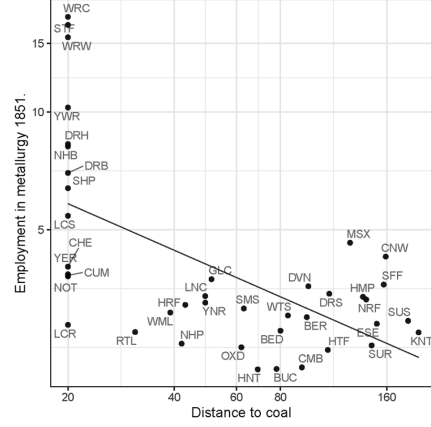


FIG. 6.—Distance to coal and employment in metallurgy 1851; logarithmic axes. It can be seen that proximity to coal was a necessary but not sufficient condition for a large industry: the three West Midlands counties around Birmingham (Stafford [STF], Warwick [WRW], and Worcester [WRC]), and West Yorkshire around Sheffield had been accumulating skill in metalworking since at least the sixteenth century. Full county names corresponding to these abbreviations can be found in the replication files.

TABLE 6
EMPLOYMENT SHARES OF TRADITIONAL INDUSTRIES IN 1851:
SEMIPARAMETRIC SPATIAL REGRESSIONS

	DEPENDENT VARIABLE: SHARE OF MEN EMPLOYED IN TRADITIONAL INDUSTRIES, 1851			
	Food	Garments	Shoes	Wood
Skills, 1790s	-.042 (.054)	.154 (.112)	.025 (.084)	.028 (.152)
Wage, 1760s	-.600 (.314)	-.652 (.654)	-.523 (.498)	.290 (.885)
Market potential, 1750	-.010 (.034)	-.021 (.070)	.028 (.051)	-.116 (.095)
Adjusted R^2	.476	.372	.728	.017
AIC	-48.1	11.9	-23.0	36.4
$S(\text{lon, lat})$	.000	.312	.692	.780

NOTE.—Dependent variable is the employment share of males aged 20–29 in 1851. Semiparametric spatial regressions: $S(\text{lon, lat})$ gives the approximate significance level of a thin-plate spline in longitude and latitude. Standard errors are in parentheses. Regressions include a dummy for London, and “Shoes” includes one for Northamptonshire. All other variables are in logs. AIC = Akaike information criterion.

- This supports a *sector-specific capability* story: mechanical skill matters for mechanized textiles/metals, not for all manufacturing.

8 Appendix A: why the spatial method matters (Tables A1–A2)

8.1 Table A1: standard-error corrections can deliver different inference

Point. If regressors and omitted determinants are spatially patterned, then inference can be sensitive to: (i) whether one controls for geographic trends, and (ii) how one tunes spatial standard-error corrections.

8.2 Table A2: Monte Carlo coverage and the case for semiparametric spatial regression

Monte Carlo experiments compare:

TABLE A1
SIGNIFICANCE LEVELS OF THE MAIN EXPLANATORY VARIABLE IN THREE REGRESSIONS
USING DIFFERENT STANDARD ERROR CORRECTIONS

DEGREE	ROBUST	CLUSTERED	CONLEY		BCH		CRS		IM		MW	
			50(0)	100(0)	4	6	8	12	8	12	.01	.05
			km	km								
Africa (<i>N</i> = 379)												
0	.000	.001	.000	.000	.084	.032	.031	.020	.174	.300	.005	.046
1	.000	.002	.000	.000	.084	.032	.125	.964	.103	.415	.004	.043
2	.000	.006	.001	.001	.168	.058	.352	.963	.256	.465	.013	.103
Germany (<i>N</i> = 324)												
0	.008	.054	.014	.034	.185	.095	.508	.613	.486	.620	.103	.109
1	.014	.047	.018	.050	.304	.108	.805	.127	.808	.178	.106	.167
2	.009	.041	.015	.055	.287	.179	.961	.340	.943	.275	.128	.228
Global (<i>N</i> = 114)												
0	.000	.032	.000	.001	.319	.190	.016	.999	.273	.717	.059	.288
1	.005	.037	.004	.006	.259	.139	.211	.692	.300	.735	.022	.123
2	.007	.001	.006	.010	.122	.068	.805	.834	.526	.952	.042	.065

NOTE.—In each case, significance levels from .01 upward may be obtained by an appropriate choice of correction and assumed tuning parameter, along with the degree of the included polynomial in longitude and latitude. Significance levels for different standard error corrections. “Degree” presents the polynomial degree of longitude and latitude variables added to each regression: no variables, linear, and quadratic. “Robust” denotes heteroskedasticity-robust standard errors. “Cluster” gives clustered standard errors when the observations were divided into 20 groups of nearest neighbors by k -means clustering. “Conley” denotes Conley (1999) standard errors with a Bartlett kernel and cutoff distances of 50 and 100 km for the German data and 500 and 1,000 km otherwise. “BCH” (Bester, Conley, and Hansen 2011), “CRS” (Canay, Romano, and Shaikh 2017), and “IM” (Ibragimov and Müller 2010) are based on large clusters, again chosen by k -means clustering, with the assumed number of clusters given below each. “MW” gives Müller and Watson (2021) values, assuming average correlation values between residuals of 0.01 and 0.05.

TABLE A2
MONTE CARLO ESTIMATES OF 95% COVERAGE PROBABILITIES FOR SEMIPARAMETRIC
REGRESSIONS AND STANDARD ERROR CORRECTIONS OVER THREE SETS
OF GEOGRAPHICAL COORDINATES

DEGREE	NO TREND				TREND			
	SP	SP2	OLS	Conley	SP	SP2	OLS	Conley
England								
0	.93	.93	.91	.88	.85	.93	.70	.66
1	.93	.93	.92	.88	.85	.93	.74	.67
2	.93	.93	.93	.89	.85	.93	.73	.65
Africa								
0	.91	.93	.64	.82	.90	.93	.41	.58
1	.91	.93	.68	.81	.90	.93	.41	.56
2	.91	.93	.72	.79	.90	.93	.45	.55
Germany								
0	.92	.94	.74	.82	.93	.94	.43	.57
1	.92	.94	.78	.82	.93	.94	.49	.57
2	.92	.94	.80	.81	.93	.94	.48	.54

NOTE.—Proportion of 95% confidence intervals containing the true coefficient of zero. Simulations are based on the coordinates of 41 English counties, 150 African tribal areas, and 100 German counties, where the variables do or do not contain spatial trends. “SP” denotes a semiparametric regression with a thin-plate spline in latitude and longitude; “SP2” uses a two-step procedure for potential spatial confounding; “OLS” denotes ordinary least squares with robust standard errors; “Conley” uses a Conley kernel correction. “Degree” presents the polynomial degree of longitude and latitude variables added to each regression: no variables, linear, and quadratic.

- OLS with different spatial standard-error fixes,
- versus semiparametric spatial regressions (SP / SP2),

and evaluate 95% confidence interval *coverage*.

Interpretation.

- When spatial trends/confounds are present, some SE fixes can yield severe under-coverage (confidence intervals too narrow) or over-coverage.
- Semiparametric controls that flexibly absorb spatial structure tend to restore coverage closer to nominal levels.

9 Short summary

- **Main conclusion.** The geography of early British industrialization is strongly associated with pre-existing mechanical/toolmaking skills, even after controlling for wages, market access, and flexible geography. Mechanical skills look like a *capability constraint* for adopting Industrial Revolution technologies.
- **Complementarities.** Low agricultural wages (comparative advantage away from agriculture) and market potential (demand/transport access) naturally complement skills. Coal is crucial for metals; it is less central for early textiles.
- **Why it matters.** This strengthens a view of industrialization as not just “high wages + coal” but also as a story about *localized human capital of a very specific type* (mechanical competence) shaped by earlier craft traditions and training costs.
- **Method lesson.** In small- N spatial cross-sections, inference can be fragile. Modeling spatial structure (not just correcting SEs) is central.

10 Conclusion

10.1 Summary

- **Main empirical conclusion.** The county-level geography of early British industrialization—especially mechanized **cotton textiles** and **metals**—is strongly and robustly associated with the local abundance of **mechanical and toolmaking skills** at the dawn of mechanization (proxied by the cohort born in the 1790s).
- **Interpretation of the core correlates.** Industrialization is not explained by “geography alone”: even after flexibly controlling for location (a semiparametric spatial trend), **mechanical skills** remain predictive, while **low agricultural wages** (a proxy for low agricultural productivity / poor soil) and **market potential** (access to demand) systematically matter as well.
- **Sector-specificity (a key falsification idea).** The same variables that explain mechanized textiles and metals do *not* systematically explain employment in **traditional** industries whose technologies changed little in this period (e.g., food, garments, shoes, wood). This supports a *capability constraint* story rather than a generic “bigger counties industrialize more” story.
- **Metals versus textiles.** For **metallurgy**, persistence/industrial tradition and coal-related energy advantages matter more strongly, while mechanical skills still complement metal production, especially in machinery- and metal-products activities. For **textiles**, coal is less central early on; skill, wages, and market access dominate.
- **A broader historical conclusion.** The Industrial Revolution should not be understood as a narrowly confined episode in a few sectors. Even if early macro effects look modest (because leading sectors started small), the paper aligns with a view of sustained improvements across many activities and a long pre-1800 trajectory of industrial growth.

10.2 Economic intuition: why “mechanics” matter for an Industrial Revolution

- **Mechanical skills are an adoption constraint.** New mechanized technologies require installation, maintenance, repair, incremental redesign, and toolmaking. These tasks rely on tacit “hands-on” competence that cannot be instantly imported or written down. Counties with thicker markets for such artisans can adopt and operate frontier machinery earlier and at lower effective cost.
- **Market integration and comparative advantage shape where skills accumulate.** As Britain becomes more integrated, some areas specialize away from agriculture. Regions with poor agricultural fundamentals (low wages) move toward manufacturing, which raises local demand for artisans, reinforcing the local stock of mechanical competence that later becomes critical for mechanization.
- **Skill formation is institutional and market-based.** Britain’s apprenticeship system is emphasized as relatively flexible and responsive to market incentives. In the data, lower bar-

riers/costs of training (proxied by watchmaking apprenticeship fees) predict more mechanical skill. This makes mechanical competence both *historically produced* and *locally persistent*.

- **Upper-tail human capital amplifies small average differences.** If mechanical competence has a distribution, small shifts in its mean can generate large differences in the thickness of the upper tail. In a technological era where top artisans and tacit know-how are pivotal, the “upper tail” can disproportionately determine success.

10.3 Methodological intuition: why the spatial approach is part of the conclusion

- **Spatial correlation is not only a standard-error problem.** When both outcomes and covariates follow geographic trends, coefficients can reflect spatial confounding (omitted geography), not just correlated residuals. The semiparametric spatial regression is used to absorb broad geographic structure directly.
- **Why this matters for interpretation.** The paper’s conclusion is not merely that “industry is clustered”; it is that **conditional on flexible geography**, mechanical skills still explain where mechanized industry takes off.

10.4 Takeaway

The paper argues that the Industrial Revolution had “mechanics” in a literal sense: local concentrations of mechanical and toolmaking skills created a capability advantage for adopting and operating mechanized technologies. Market access and low agricultural productivity helped push regions into manufacturing, but only places with sufficiently thick artisan competence could translate these forces into mechanized industrialization. The result is a path-dependent geography of industrial takeoff driven by upper-tail, tacit human capital rather than broad education, finance, or geography alone.